$$u = a(x^2 + y^2)$$
$$v = bxy$$
$$w = 0$$

10)

$$E_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

$$\begin{bmatrix} E_{11} & E_{12} & E_{13} \\ & E_{22} & E_{23} \\ \text{Simétrico} & & E_{33} \end{bmatrix}$$

$$E_{11} = \frac{\partial u_1}{\partial x_1} = \overset{2xa}{\frac{\partial u}{\partial x}} \quad , \quad E_{22} = \frac{\partial u_2}{\partial x_2} = \overset{bx}{\frac{\partial v}{\partial y}} \quad , \quad E_{33} = \frac{\partial u_3}{\partial x_3} = \overset{0}{\frac{\partial w}{\partial z}}$$

$$E_{12} = E_{21} = \frac{1}{2}\left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}\right) = \frac{1}{2}\left(\overset{2ya}{\frac{\partial u}{\partial y}} + \overset{by}{\frac{\partial v}{\partial x}}\right)$$

$$E_{13} = E_{31} = \frac{1}{2}\left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1}\right) = \frac{1}{2}\left(\overset{0}{\frac{\partial u}{\partial z}} + \overset{0}{\frac{\partial w}{\partial x}}\right)$$

$$E_{23} = E_{32} = \frac{1}{2}\left(\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2}\right) = \frac{1}{2}\left(\overset{0}{\frac{\partial v}{\partial z}} + \overset{0}{\frac{\partial w}{\partial y}}\right)$$

$$E(x,y) = \begin{bmatrix} 2xa & \frac{1}{2}(2ya+by) & 0 \\ \cdot & bx & 0 \\ & & 0 \end{bmatrix}$$